

to threats at the endpoint level, providing an additional layer of protection.

A five-step journey to zero trust

A zero trust security model (ZTNA) is gaining prominence in the IoT security landscape. To implement a zero trust strategy effectively, a systematic approach is essential. To implement it effectively, consider these three core principles:

Least privileged access. Limit access rights to the bare minimum required for an individual's role. Regularly review and adjust permissions to reduce the risk of unauthorised access.

Continuous trust verification. Implement two-factor authentication and regularly re-authenticate users to prevent unauthorised access. Compartmentalise access to different systems to prevent lateral movement by attackers.

Strict network segmentation. Isolate different IoT components to limit the potential impact of a compromise. Each component should have its access controls.

Define production services by recognising various environments and compartmentalising assets. Delve into transaction flows within your IoT ecosystem, for a comprehensive understanding and protection of communication layers. Conduct a thorough analysis of data flows and interdependencies, carefully evaluating the connections between IoT applications to pinpoint potential vulnerabilities. In the quest for robust security, it is imperative to build the zero trust network, establishing zones and assessing the potential impact of security breaches, and subsequently designing appropriate security measures. Finally, as a critical layer of defence, define zero trust policies that answer the fundamental questions of who can access what, when, where, why, and how, ensuring that your policies align seamlessly with these considerations to fortify security.

IoT security is an ongoing journey, not a destination. Threats evolve, and vulnerabilities emerge. Regularly monitor and update your security measures to adapt to changing circumstances. By implementing a zero trust security model, leveraging automation, and conducting continuous security inspection, organisations can significantly enhance their IoT security posture. Additionally, partnering with managed security services providers (MSSPs) can provide valuable expertise and support in safeguarding IoT environments. Remember, in the world of IoT, security is a collective responsibility, and vigilance is the key to success. **EFY**

This article is from a tech talk session at EFY Expo 2023, Pune, by Prashasth Bantwal Baliga, Principal Security Consultant (Customer success), Palo Alto Networks. It has been transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Journalist at EFY.



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